

# Tab 1

# Dhruv Patel

(747) 257-7194 — [dhruv.patel.imp@gmail.com](mailto:dhruv.patel.imp@gmail.com) — [Portfolio Website](#) — [GitHub](#) — [LinkedIn](#)

## EDUCATION

---

### University of California, San Diego

B.S. in Data Science

Expected June 2026

Relevant Coursework: Statistical Inference, Machine Learning, Systems for Scalable Analytics, Data Visualization, Linear Algebra, Probability & Computing, Optimization

## TECHNICAL SKILLS

---

**Programming:** Python, R, SQL, Java, C++, JavaScript

**Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, PyTorch, scikit-learn, GeoPandas, Folium

**Analytics & Modeling:** Predictive Modeling, Hypothesis Testing, A/B Testing, Feature Engineering, Model Evaluation (AUC, Precision/Recall), Error Analysis

**Data & ML Systems:** ETL Pipelines, Dataset Validation, Model Training & Inference Pipelines

**Tools:** Git, Jupyter, Tableau, Excel, ArcGIS, AWS (basic)

**Soft Skills:** Analytical Problem Solving, Technical Communication, Cross-Functional Collaboration

## PROJECTS

---

### MNIST Neural Network – 10 Digit Classification (2025)

Implemented neural networks from scratch to classify handwritten digits (0–9), focusing on architecture design and training dynamics. Built forward and backward propagation pipelines, loss computation, and gradient-based optimization. Analyzed effects of network depth, learning rate, and activation functions on convergence and accuracy. Evaluated model performance using accuracy and error analysis, strengthening understanding of core ML principles underlying modern deep learning and transformers.

### FashionMNIST Model Comparison Dashboard (2025)

Developed an interactive Streamlit dashboard comparing MLP vs CNN models trained on FashionMNIST. Visualized accuracy, confidence scores, confusion matrices, and misclassifications to support model benchmarking. Demonstrated structured evaluation workflows directly transferable to LLM benchmarking and FM-Eval-style evaluation.

### Differential Privacy – Synthetic Telemetry Evaluation (2025)

Capstone project evaluating user-level differential privacy on synthetic telemetry data. Implemented Gaussian and Laplace mechanisms and measured privacy-utility tradeoffs using AUC,  $L_\infty$  error, and IOU metrics across 100 randomized runs. Gained hands-on experience with responsible AI, data governance constraints, and privacy-preserving ML.

### Sorting Algorithm Visualization (2025)

Built interactive visual tools to explain algorithm behavior step-by-step to learners. Emphasized cognitive accessibility, pacing, and visual clarity for users with varying technical backgrounds.

### San Diego Transit Accessibility Analysis (2025)

Conducted a GIS-based study identifying transit deserts and accessibility gaps across San Diego County. Designed visual and narrative outputs to support policy and user-centered decision-making. Published an interactive StoryMap to communicate findings to non-technical stakeholders

## WORK EXPERIENCE

---

### Peer Mentor, OASIS Summer Bridge – UC San Diego

Apr 2023 – March 2025 — San Diego, CA

- Mentored incoming first-year students, facilitated workshops, and connected them to campus academic resources.

- Contributed to increasing student retention and community integration through personalized support.

**STEM Tutor Intern, Boys and Girls Club – LA Harbor**

Mar 2022 – Aug 2022 — Los Angeles, CA

- Provided one-on-one tutoring for middle and high school students in math and science.
- Created individualized learning plans, leading to measurable improvements in student performance.